

Ref: <https://www.networkacademy.io/ccie-enterprise/multicast/troubleshooting-pim-dense-mode>

In this lab, a router's image is used on hosts, making them act like hosts!

ena conf t ho SRC no ip domain-lo int e0/0 ip add 10.1.1.1 255.255.255.0 no sh exit ip multicast-routing router ospf 1 network 0.0.0.0 0.0.0.0 area 0 end wri	ena conf t ho PC1 no ip domain-lo int e0/0 ip add 10.10.10.1 255.255.255.0 no sh exit ip multicast-routing router ospf 1 network 0.0.0.0 0.0.0.0 area 0 end wri	ena conf t ho PC2 no ip domain-lo int e0/0 ip add 10.10.20.1 255.255.255.0 no sh exit ip multicast-routing router ospf 1 network 0.0.0.0 0.0.0.0 area 0 end wri
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Devices

ena conf t ho R1 no ip domain-lo int e0/0 ip add 10.1.1.252 255.255.255.0 no sh int e0/1 ip add 10.1.2.1 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri	ena conf t ho R2 no ip domain-lo int e0/0 ip add 10.1.1.253 255.255.255.0 no sh int e0/1 ip add 10.1.2.2 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri	ena conf t ho R3 no ip domain-lo int e0/3 ip add 10.1.3.1 255.255.255.0 no sh int e0/2 ip add 10.1.4.1 255.255.255.0 no sh int e0/1 ip add 10.1.2.3 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri	ena conf t ho R4 no ip domain-lo int e0/2 ip add 10.1.6.1 255.255.255.0 no sh int e1/0 ip add 10.10.10.254 255.255.255.0 no sh int e0/1 ip add 10.1.2.4 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri
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<pre> ena conf t ho R5 no ip domain-lo int e0/2 ip add 10.1.4.2 255.255.255.0 no sh int e0/0 ip add 10.10.7.1 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri </pre>	<pre> ena conf t ho R6 no ip domain-lo int e0/0 ip add 10.1.7.2 255.255.255.0 no sh int e0/3 ip add 10.1.3.2 255.255.255.0 no sh int e0/2 ip add 10.1.6.2 255.255.255.0 no sh int e0/1 ip add 10.1.8.1 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri </pre>	<pre> ena conf t ho R7 no ip domain-lo int e0/1 ip add 10.1.8.2 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri </pre>	<pre> ena conf t ho R8 no ip domain-lo int e0/1 ip add 10.1.8.3 255.255.255.0 no sh int e1/0 ip add 10.10.20.254 255.255.255.0 no sh exit router ospf 1 network 0.0.0.0 0.0.0.0 area 0 exit ip multicast-routing end wri </pre>
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1. Ensure multicast routing is enabled on **all** devices.
2. Enable dense mode on interfaces.
3. **Configure** hosts to join **the** group 239.1.1.1 **on** all hosts.

Start Lab!

Summary Command:

- show ip igmp groups
- show ip mroute
- mtrace (destination-source-multicast)